TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Wet Test for Basic Radicals

The systematic group analysis of cations — the group reagents, why the order of separation matters, flame tests, and the confirmatory tests that name each basic radical.

CHE-1106 · CHEMISTRY-I SESSIONAL · EXPERIMENT 2 · LEVEL 1 TERM 1

1. Aim and principle

Aim: to detect the basic radical (cation) in a given salt by wet tests. Wet tests work on the **original solution** of the salt and rest on one principle: cations fall into families (**groups**) whose members share a reagent that precipitates them together while leaving later groups in solution. Precipitate group by group, in the fixed order, and a mixture untangles itself; disturb the order and later reagents precipitate earlier groups too, and the analysis collapses. The chemistry underneath is solubility product and common-ion control — the group reagents are dilute or buffered precisely so that only the intended, least-soluble family falls.

2. The group table

GROUP	GROUP REAGENT	RADICALS	PRECIPITATE
I	Dilute HCl	Pb^{2+}, Ag^{+}, Hg_2^{2+}	White chlorides
II	H_2S in presence of dilute HCl	Pb^{2+}, Cu^{2+}, Cd^{2+}, Hg^{2+}, $Sn^{2+/4+}$, Sb^{3+}…	Coloured sulphides
IIIA	$NH_4Cl + NH_4OH$	Fe^{3+}, Al^{3+}, Cr^{3+}	Hydroxides (brown / white / green)
IIIB	H_2S in presence of NH_4OH	Zn^{2+}, Mn^{2+}, Co^{2+}, Ni^{2+}	Sulphides (white / flesh / black / black)
IV	$(NH_4)_2CO_3$ in presence of $NH_4OH + NH_4Cl$	Ca^{2+}, Sr^{2+}, Ba^{2+}	White carbonates
V	No group reagent — tested individually in the filtrate	Mg^{2+}, Na^{+}, K^{+}, NH_4^{+}	— (NH_4^{+} is tested in the original salt before anything else)

Two procedural laws follow from the table: **NH_4^+ is tested first**, in the original salt (warm with NaOH — ammonia gas), because ammonium salts are added as reagents throughout and would fake the result later; and **each group's precipitate is filtered off completely** before the next reagent is added — separation (Experiment 4) is this experiment performed with full rigour.

3. Preliminary wet observations

- **Original solution with NaOH (dropwise, then excess):** white precipitate dissolving in excess → Zn^{2+}, Al^{3+} or Pb^{2+} (amphoteric hydroxides); white precipitate not dissolving → Ca^{2+}/Mg^{2+}/Ba^{2+}; blue precipitate → Cu^{2+}; brown → Fe^{3+}; green → Ni^{2+}/Cr^{3+}/Fe^{2+}; ammonia smell on warming → NH_4^+.
- **With NH_4OH (dropwise, then excess):** deep blue solution → Cu^{2+} (the tetraammine complex, Chemistry-I Topic 4 in a test tube); white precipitate dissolving → Zn^{2+}; brown → Fe^{3+}; pale blue → Ni^{2+}.
- **Flame test** (clean platinum/nichrome wire, concentrated HCl, into the non-luminous flame): golden yellow → Na^+; lilac (viewed through cobalt glass to mask sodium) → K^+; brick red → Ca^{2+}; crimson → Sr^{2+}; apple green → Ba^{2+}; blue-green → Cu^{2+}.

4. Confirmatory tests — the commonly met radicals

RADICAL	CONFIRMATORY TEST AND POSITIVE RESULT
Pb^{2+}	KI gives a bright yellow precipitate (PbI_2) that dissolves on heating and reappears as golden spangles on cooling; K_2CrO_4 also gives yellow $PbCrO_4$.
Cu^{2+}	NH_4OH in excess gives the deep blue tetraammine solution; potassium ferrocyanide gives a chocolate-brown precipitate.
Fe^{3+}	Potassium thiocyanate gives a blood-red colour; potassium ferrocyanide gives deep (Prussian) blue.
Al^{3+}	NaOH: white gelatinous precipitate dissolving in excess; reprecipitated by NH_4Cl on boiling. (Lake test with aluminon gives a red lake.)
Zn^{2+}	NaOH white precipitate soluble in excess; potassium ferrocyanide gives a white/cream precipitate; H_2S in alkaline solution gives white ZnS.
Ca^{2+}	Ammonium oxalate gives a white precipitate (CaC_2O_4) insoluble in acetic acid; flame brick red.
Ba^{2+}	K_2CrO_4 gives a yellow precipitate ($BaCrO_4$) insoluble in acetic acid; flame apple green.

Mg^{2+}	In presence of NH_4Cl/NH_4OH, disodium hydrogen phosphate gives a white crystalline precipitate ($MgNH_4PO_4$) on standing/scratching.
NH_4^+	Warm with NaOH: ammonia gas (smell; turns red litmus blue; white fumes with HCl rod); Nessler's reagent gives a brown colour/precipitate.
Na^+ / K^+	By flame (golden yellow persistent / lilac through cobalt glass); K^+ also: sodium cobaltinitrite gives a yellow precipitate.

5. Precautions

- Prepare the original solution correctly: water first; if insoluble, dilute HCl or HNO_3 as the analysis directs — never concentrated acids casually.
- Add group reagents dropwise with shaking; excess H_2S water must be fresh (it oxidises in air).
- Wash precipitates before confirmatory tests so mother liquor does not fake colours.
- Clean the flame-test wire in concentrated HCl until it colours the flame no more.

6. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why is NH_4^+ tested in the original salt before the group analysis begins?
2. A solution gives a white precipitate with NaOH that dissolves in excess, and a white precipitate with H_2S in alkaline medium. Name the radical and the group it belongs to.
3. State the group reagents of Groups I, IIIA and IV, and one radical of each group.
4. How do you distinguish Ca^{2+} from Ba^{2+} using (a) a flame, (b) one wet reagent?
5. Why does the deep blue colour with excess NH_4OH confirm Cu^{2+} rather than merely suggest it?
6. A golden-yellow flame persists and masks everything. Which radical is present, and how is K^+ looked for in its presence?
7. Give the confirmatory test for Fe^{3+} with thiocyanate and for Pb^{2+} with KI (including the hot/cold behaviour).

Answer key

1. Because NH_4Cl and NH_4OH are used as reagents in Groups III–IV; any ammonium found later could have come from the reagents. Only a test on the untouched original salt can prove the salt itself contained NH_4^+.
2. Zn^{2+} (amphoteric hydroxide dissolving in excess NaOH; white ZnS in alkaline H_2S) — Group IIIB.

3. Group I: dilute HCl (Pb^{2+}). Group IIIA: NH_4Cl + NH_4OH (Fe^{3+}). Group IV: $(NH_4)_2CO_3$ in presence of NH_4Cl + NH_4OH (Ca^{2+}). Any one radical of each earns the mark.
4. (a) Flame: Ca^{2+} brick red; Ba^{2+} apple green. (b) K_2CrO_4: Ba^{2+} gives a yellow precipitate insoluble in acetic acid; Ca^{2+} gives no such precipitate under these conditions. (Ammonium oxalate precipitates Ca^{2+} white — the converse distinction.)
5. Because the deep blue is a specific chemical species — the tetraamminecopper(II) complex $[Cu(NH_3)_4]^{2+}$ — formed by Cu^{2+} under these conditions and by no other radical in the scheme; colour here is structure, not coincidence.
6. Na^+ (persistent golden yellow). K^+ is looked for through cobalt-blue glass, which absorbs the sodium yellow and lets the lilac potassium flame show.
7. Fe^{3+} + KCNS → blood-red colour. Pb^{2+} + KI → bright yellow PbI_2, dissolving on warming and crystallising on cooling as golden spangles.
